VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Artificial Intelligence (23CS5PCAIN)

Submitted by

B Prabhanjan(1BM23CS060)

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Artificial Intelligence (23CS5PCAIN)" carried out by **B Prabhanjan(1BM23CS060),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Artificial Intelligence (23CS5PCAIN) work prescribed for the said degree.

Swathi Sridharan Assistant Professor Department of CSE, BMSCE	Dr. Kavitha Sooda Professor & HOD Department of CSE, BMSCE

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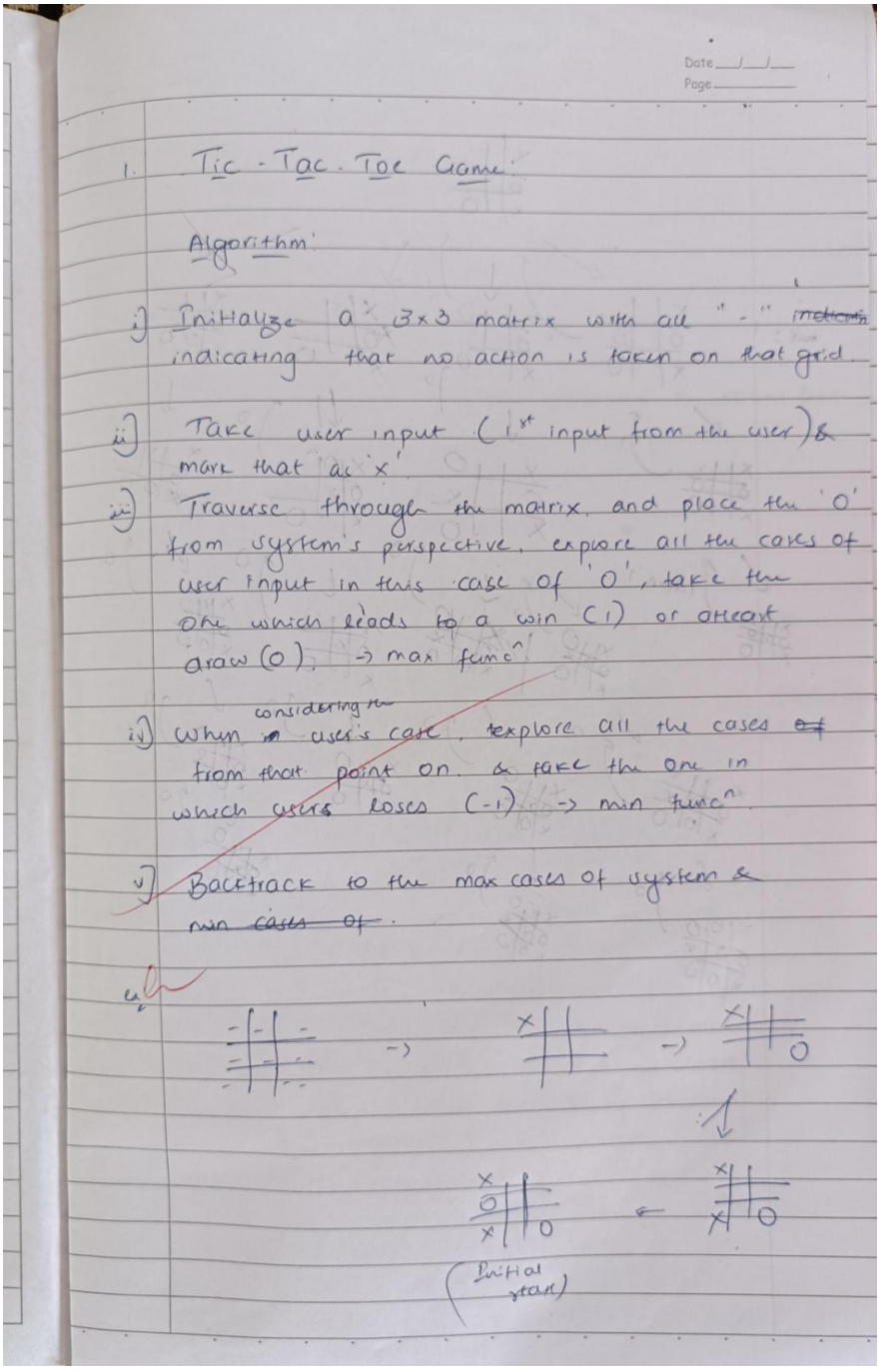
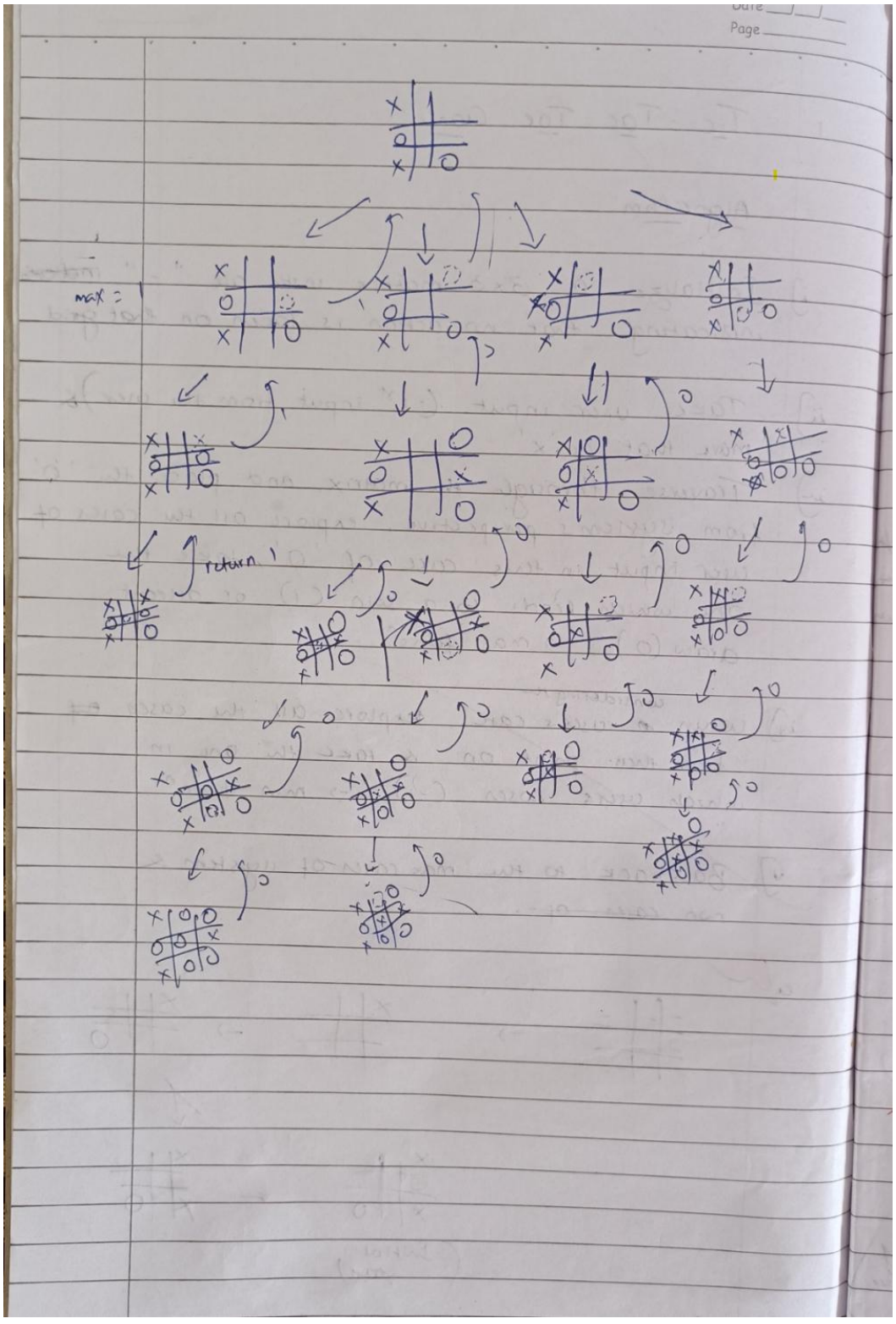
Github Link:
https://github.com/Prabhanjann/AI

Program 1

a) Implement Tic –Tac –Toe Game
b) Implement vacuum cleaner agent

Algorithm:




max = 1
return 1

Date: / /
Page:

Vacuum cleaner:

Algorithm:

→ Set up the room

→ Place the agent, starts cleaning the room from a random dirty spot.

→ Start the cleaning process.

→ Repeat until the whole room is cleaned:
- the agent looks / scans the whole room repeatedly
- cleans the spot which is dirty ~~by [illegible]~~
- If the spot is clean, move to the next.

→ Stop the agent.

~~[illegible]~~:
Start at spot B.
Spot B is dirty → clean
Move to spot A → already clean → skip
Move to ~~room~~ spot C → dirty → clean
Move to spot D → dirty → clean
All spots covered → stop

input:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow \text{skip } (0,0) \xrightarrow{\text{clean}} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \xrightarrow{\text{clean}} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \leftarrow \text{skip}(1,1)$$

1 → dirty
0 → clean

Code:

a)import math

```
# Print the board
def print_board(board):
    for row in board:
        print(" | ".join(row))
        print("-" * 5)

# Check winner
def check_winner(board):
    # Rows
    for row in board:
        if row[0] == row[1] == row[2] != " ":
            return row[0]
    # Columns
    for col in range(3):
        if board[0][col] == board[1][col] == board[2][col] != " ":
            return board[0][col]
    # Diagonals
    if board[0][0] == board[1][1] == board[2][2] != " ":
        return board[0][0]
    if board[0][2] == board[1][1] == board[2][0] != " ":
        return board[0][2]
    return None

# Check if moves left
def is_moves_left(board):
    for row in board:
        if " " in row:
            return True
    return False

# Minimax function
def minimax(board, depth, isMax):
    winner = check_winner(board)
    if winner == "X":  # Computer
        return 10 - depth
    if winner == "O":  # Human
        return depth - 10
    if not is_moves_left(board):
        return 0

    if isMax:  # Computer turn
        best = -math.inf
        for i in range(3):
```

```
            for j in range(3):
                if board[i][j] == " ":
                    board[i][j] = "X"
                    best = max(best, minimax(board, depth + 1, False))
                    board[i][j] = " "
        return best
    else:  # Human turn
        best = math.inf
        for i in range(3):
            for j in range(3):
                if board[i][j] == " ":
                    board[i][j] = "O"
                    best = min(best, minimax(board, depth + 1, True))
                    board[i][j] = " "
        return best

# Find best move for computer
def best_move(board):
    bestVal = -math.inf
    move = (-1, -1)
    for i in range(3):
        for j in range(3):
            if board[i][j] == " ":
                board[i][j] = "X"
                moveVal = minimax(board, 0, False)
                board[i][j] = " "
                if moveVal > bestVal:
                    move = (i, j)
                    bestVal = moveVal
    return move

# Game loop
def play_game():
    board = [[" " for _ in range(3)] for _ in range(3)]
    human = "O"
    computer = "X"

    print("Tic-Tac-Toe Game!")
    print_board(board)

    while True:
        # Human move
        row, col = map(int, input("Enter row and col (0-2): ").split())
        if board[row][col] != " ":
            print("Cell already taken!")
            continue
        board[row][col] = human
```

```
        if check_winner(board) == human:
            print_board(board)
            print("You win!")
            break
        if not is_moves_left(board):
            print_board(board)
            print("It's a Draw!")
            break
        # Computer move
        print("Computer is thinking...")
        r, c = best_move(board)
        board[r][c] = computer

        print_board(board)
        if check_winner(board) == computer:
            print("Computer wins!")
            break
        if not is_moves_left(board):
            print("It's a Draw!")
            break

play_game()
b)class VacuumCleaner:
    def __init__(self, room_A, room_B, vacuum_pos):
        self.room_A = room_A
        self.room_B = room_B
        self.vacuum_pos = vacuum_pos

    def show_status(self):
        print(f"Vacuum in Room {self.vacuum_pos} | Room A: {self.room_A}, Room B:
{self.room_B}")

    def clean(self):
        print("Initial State:")
        self.show_status()
        print("------")

        steps = 0
        while self.room_A == "Dirty" or self.room_B == "Dirty":
            steps += 1
            print(f"Step {steps}:")

            if self.vacuum_pos == "A":
                if self.room_A == "Dirty":
                    print("Room A is Dirty → Cleaned!")
                    self.room_A = "Clean"
```

```
            else:
                print("Room A is Clean → Move to Room B")
                self.vacuum_pos = "B"

        elif self.vacuum_pos == "B":
            if self.room_B == "Dirty":
                print("Room B is Dirty → Cleaned!")
                self.room_B = "Clean"
            else:
                print("Room B is Clean → Move to Room A")
                self.vacuum_pos = "A"

        self.show_status()
        print("------")

    print("Goal Reached! Both rooms are clean.")
# ---- Main Program ----
room_A = input("Enter status of Room A (Clean/Dirty): ").capitalize()
room_B = input("Enter status of Room B (Clean/Dirty): ").capitalize()
vacuum_pos = input("Enter initial position of Vacuum (A/B): ").upper()

agent = VacuumCleaner(room_A, room_B, vacuum_pos)
agent.clean()
```

Output:

a)

```
Enter row and col (0-2): 0 0
Computer is thinking...
O |   |
-----
  | X |
-----
  |   |
-----
Enter row and col (0-2): 2 1
Computer is thinking...
O |   |
-----
X | X |
-----
  | O |
-----
Enter row and col (0-2): 2 2
Computer is thinking...
O |   |
-----
X | X | X
-----
  | O | O
-----
Computer wins!
```

b)

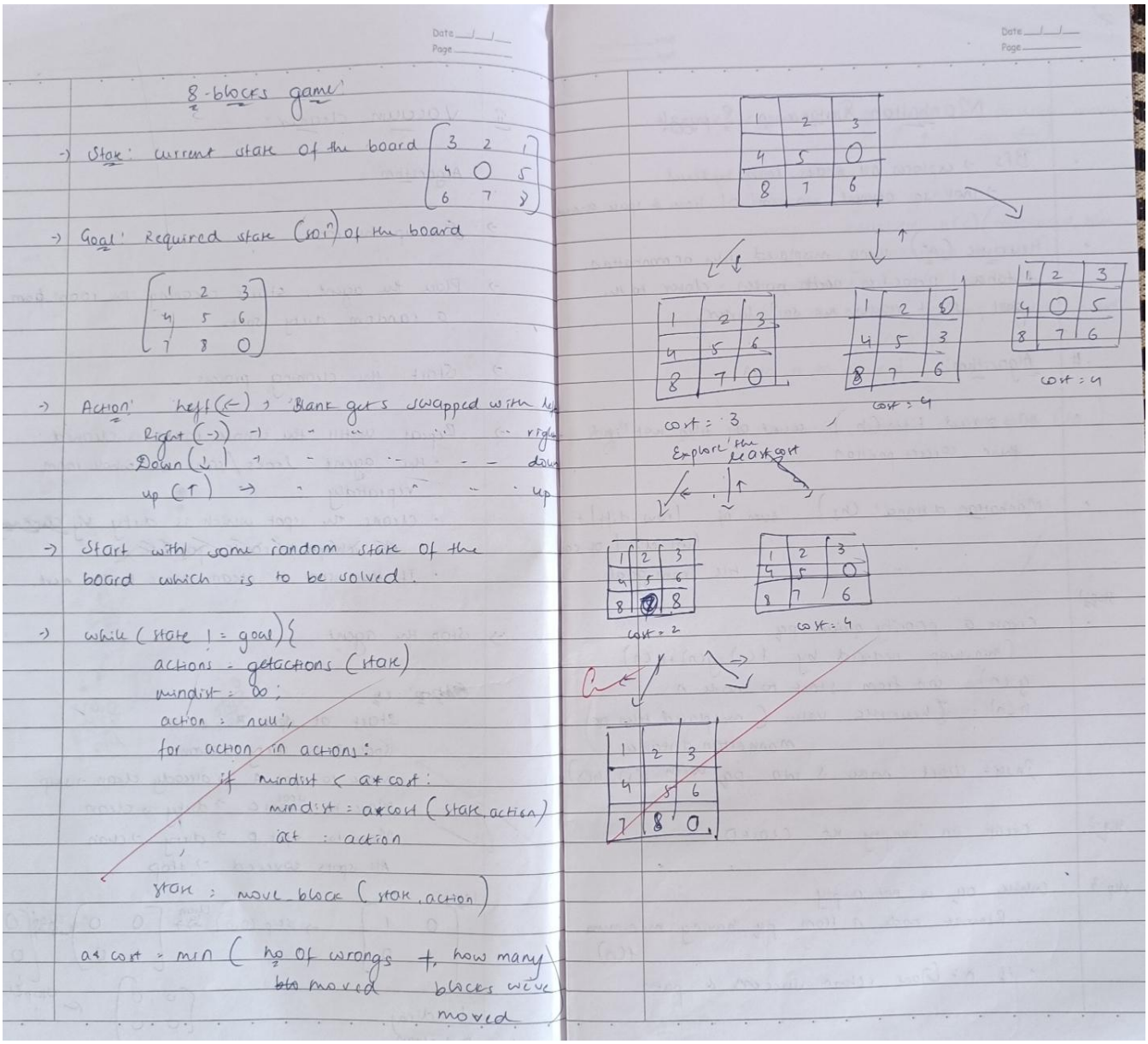
```
Enter status of Room A (Clean/Dirty): Dirty
Enter status of Room B (Clean/Dirty): Clean
Enter initial position of Vacuum (A/B): A
Initial State:
Vacuum in Room A | Room A: Dirty, Room B: Clean
------
Step 1:
Room A is Dirty → Cleaned!
Vacuum in Room A | Room A: Clean, Room B: Clean
------
Goal Reached! Both rooms are clean.
```

Program 2:

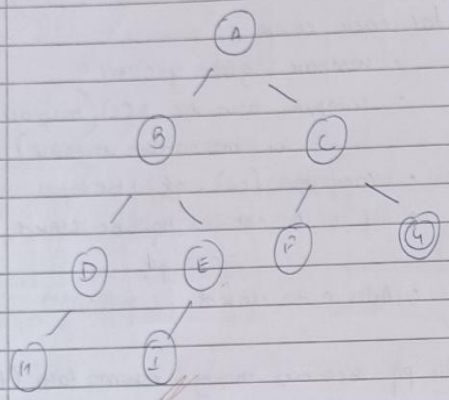
a)Implement 8 puzzle problems using Depth First Search (DFS)

b)Implement Iterative deepening search algorithm

Algorithm:



IDDFS



Iteration - 0 : A
Iteration - 1 : A B C
Iteration - 2 : A B D E C F G → Finish

Algorithm:

i) Define the graph
ii) start from root/source node
iii) Explore the level in which root/src is present. After this traversal, go to the next level
iv) Traverse the reached level & repeat the above step till you reach the goal/end state of the graph
v) if goal is found, return true
else, return false.

8 puzzle using IDDFS:

Initial state: (1,2,3)
(4,0,5)
(7,8,6)

Iteration-1 : Depth limit = 0
Move 0 right (0 up) Move 0 left (0 do

(1,2,3) (1,0,3) (1,2,3) (1,2,3
(4,5,0) (4,2,5) (0,4,5) (4,8,5
(7,8,6) (7,8,6) (7,8,6) (7,0,

Iteration-2
Move 0 right left up Down

(1,2,3)
(4,5,6)
(7,8,0)
Goal state

(1,2,3) (1,2,0)
(4,0,5) (4,5,3)
(7,8,6) (7,8,6)

Code:

a)

```
import collections

# Define the GOAL_STATE for the 8-puzzle.
# This serves as a concrete example of a "state" in a search problem.
GOAL_STATE = (
    (1, 2, 3),
    (4, 5, 6),
    (7, 8, 0)  # 0 represents the blank space
)

def find_blank(state):
    """
    Helper function to find the coordinates of the blank space (0) in a state.
    """
    for r in range(3):
        for c in range(3):
            if state[r][c] == 0:
                return r, c
    return None

def get_neighbors(state):
    """
    Generates all valid neighbor states from the current state.

    In a search problem, this function defines the "transitions" between states.
    For the 8-puzzle, this means moving the blank tile.

    Args:
        state (tuple): The current puzzle state.

    Returns:
        list: A list of new, valid puzzle states (tuples).
    """
    r_blank, c_blank = find_blank(state)
    neighbors = []

    # Possible moves for the blank space: (dr, dc) -> Up, Down, Left, Right
    moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]

    for dr, dc in moves:
        new_r, new_c = r_blank + dr, c_blank + dc

        # Check if the new position is within the grid boundaries
        if 0 <= new_r < 3 and 0 <= new_c < 3:
            # Create a mutable copy of the state to make a change
```

```
            new_state_list = [list(row) for row in state]

            # Swap the blank space with the tile at the new position
            new_state_list[r_blank][c_blank], new_state_list[new_r][new_c] = \
                new_state_list[new_r][new_c], new_state_list[r_blank][c_blank]

            # Convert the list back to a tuple for hashability
            new_state_tuple = tuple(tuple(row) for row in new_state_list)
            neighbors.append(new_state_tuple)

    return neighbors

def dfs_with_states(initial_state):
    """
    Performs a Depth-First Search on a state-space problem.

    This function explicitly uses a stack to manage states and a set to
    track visited states, demonstrating the core concepts of DFS.

    Args:
        initial_state (tuple): The starting configuration of the puzzle.

    Returns:
        list: The path to the goal state if found, or None.
    """
    # The stack will store tuples of (state, path_to_state)
    stack = collections.deque([(initial_state, [initial_state])])

    # The set stores visited states to prevent cycles and redundant work.
    visited = {initial_state}

    while stack:
        current_state, path = stack.pop()

        # Check if the current state is the goal state
        if current_state == GOAL_STATE:
            return path

        # Explore the neighbors of the current state
        # The order of neighbors determines the search path (depth-first).
        for neighbor in get_neighbors(current_state):
            # If the neighbor has not been visited, add it to the stack
            if neighbor not in visited:
                visited.add(neighbor)
                new_path = path + [neighbor]
                stack.append((neighbor, new_path))
```

```
    return None # No solution found

def print_path(path):
    """
    Prints the sequence of states in the solution path.
    """
    if path is None:
        print("No solution found.")
        return

    for i, state in enumerate(path):
        print(f"--- Step {i} ---")
        for row in state:
            print(" ".join(map(str, row)))
    print(f"--- Goal Reached in {len(path)-1} steps ---")

# --- Example Usage ---
if __name__ == "__main__":
    initial_state = (
        (1, 2, 3),
        (4, 5, 6),
        (7, 0, 8)
    )

    print("Solving the 8-puzzle using DFS with explicit states...")
    solution_path = dfs_with_states(initial_state)

    if solution_path:
        print_path(solution_path)
    else:
        print("No solution found.")

b)
import collections

# Define the goal state
GOAL_STATE = (
    (1, 2, 3),
    (4, 5, 6),
    (7, 8, 0)
)

def find_blank(state):
    """Finds the coordinates of the blank space."""
    for r in range(3):
        for c in range(3):
          if state[r][c] == 0:
```

```
                return r, c
    return None

def get_neighbors(state):
    """Generates all valid neighbor states."""
    r_blank, c_blank = find_blank(state)
    neighbors = []
    moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]
    for dr, dc in moves:
        new_r, new_c = r_blank + dr, c_blank + dc
        if 0 <= new_r < 3 and 0 <= new_c < 3:
            new_state_list = [list(row) for row in state]
            new_state_list[r_blank][c_blank], new_state_list[new_r][new_c] = \
                new_state_list[new_r][new_c], new_state_list[r_blank][c_blank]
            new_state_tuple = tuple(tuple(row) for row in new_state_list)
            neighbors.append(new_state_tuple)
    return neighbors

def depth_limited_search(initial_state, limit):
    """
    The DLS function is a core component of IDS.
    """
    stack = collections.deque([(initial_state, [initial_state])])
    visited = {initial_state}

    while stack:
        current_state, path = stack.pop()

        if current_state == GOAL_STATE:
            return path

        if len(path) > limit:
            continue

        for neighbor in get_neighbors(current_state):
            if neighbor not in visited:
                visited.add(neighbor)
                new_path = path + [neighbor]
                stack.append((neighbor, new_path))

    return None

def iterative_deepening_search(initial_state):
    """
    Solves the 8-puzzle using Iterative Deepening Search (IDS).

    IDS performs a series of DLS searches, with the depth limit increasing
```

```
    with each iteration.

    Args:
        initial_state (tuple): The starting configuration.

    Returns:
        list: The path to the goal state if a solution exists.
    """
    # The depth limit starts at 0 and increases with each iteration.
    depth = 0
    # A safe upper bound for the depth to prevent infinite loops on unsolvable puzzles
    # For a 3x3 puzzle, the maximum path length is about 31 steps.
    max_depth = 50

    while depth < max_depth:
        print(f"Trying depth limit: {depth}")
        solution = depth_limited_search(initial_state, limit=depth)

        if solution:
            return solution

        # Increment the depth limit for the next iteration
        depth += 1

    return None # No solution found within the maximum depth

def print_path(path):
    """Prints the sequence of states in the solution path."""
    if path is None:
        print("No solution found.")
        return

    for i, state in enumerate(path):
        print(f"--- Step {i} ---")
        for row in state:
            print(" ".join(map(str, row)))
    print(f"--- Goal Reached in {len(path)-1} steps ---")

# --- Example Usage ---
if __name__ == "__main__":
    initial_state = (
        (1, 2, 3),
        (4, 5, 6),
        (7, 0, 8)
    )

    print("Solving the 8-puzzle with IDS...")
```

```
solution_path = iterative_deepening_search(initial_state)

if solution_path:
    print_path(solution_path)
else:
    print("Could not find a solution.")
```

Output:

a)

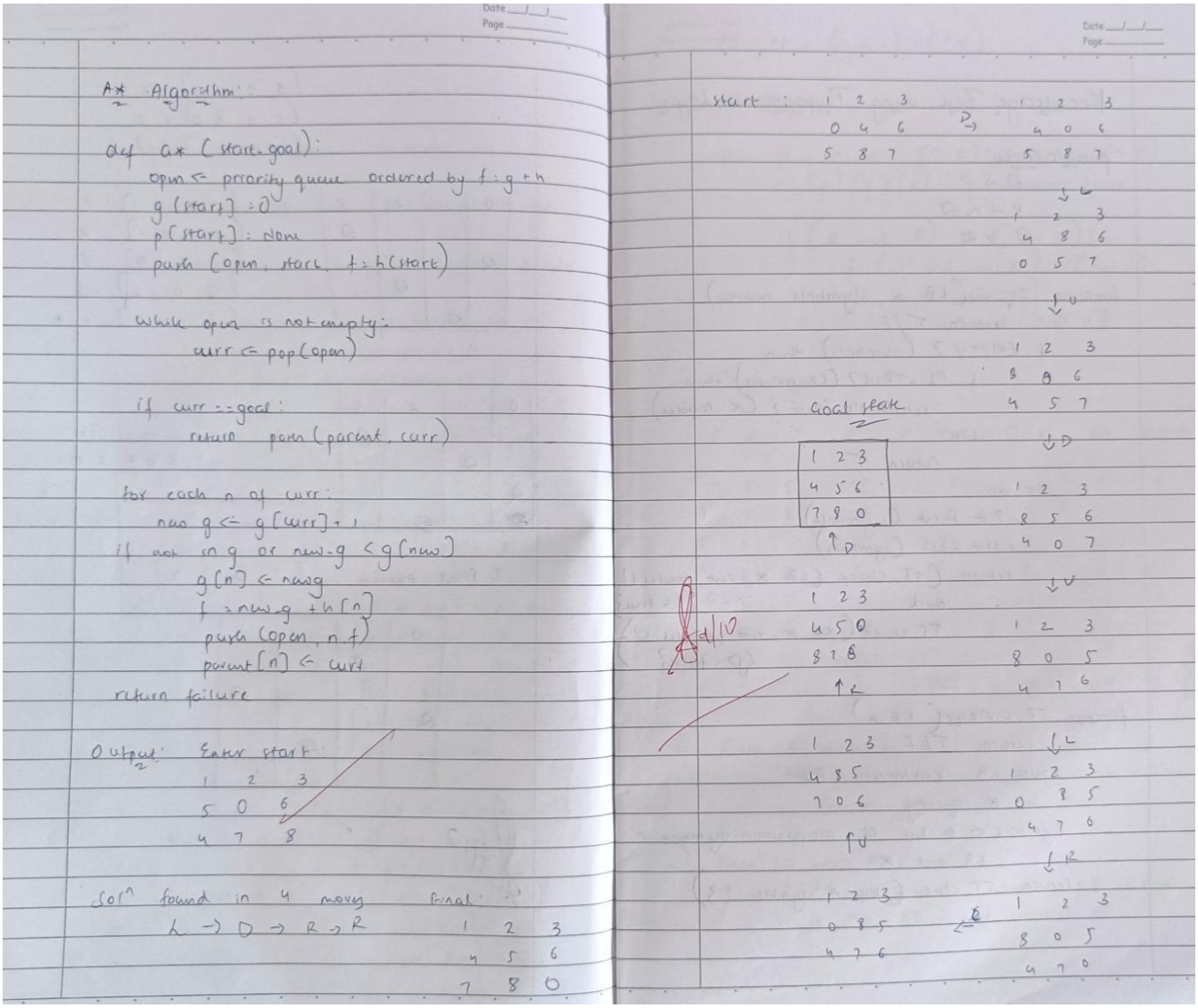
```
Solving the 8-puzzle using DFS with explicit states...
--- Step 0 ---
1 2 3
4 5 6
7 0 8
--- Step 1 ---
1 2 3
4 5 6
7 8 0
--- Goal Reached in 1 steps ---
```

b)

```
Solving the 8-puzzle with IDS...
Trying depth limit: 0
Trying depth limit: 1
--- Step 0 ---
1 2 3
4 5 6
7 0 8
--- Step 1 ---
1 2 3
4 5 6
7 8 0
--- Goal Reached in 1 steps ---
```

Program 3:
Implement A* search algorithm

Algorithm:



Code:

```
import heapq

class PuzzleState:
    def __init__(self, board, parent=None, move="", depth=0, cost=0):
        self.board = board
        self.parent = parent
        self.move = move
        self.depth = depth
        self.cost = cost

    def __lt__(self, other):
        return self.cost < other.cost

    def find_blank(self):
        return self.board.index(0)

    def get_moves(self):
        blank = self.find_blank()
        moves = []
        row, col = divmod(blank, 3)
        directions = {
            "Up": (row - 1, col),
            "Down": (row + 1, col),
            "Left": (row, col - 1),
            "Right": (row, col + 1)
        }
        for move, (r, c) in directions.items():
            if 0 <= r < 3 and 0 <= c < 3:
                new_blank = r * 3 + c
                new_board = self.board[:]
                new_board[blank], new_board[new_blank] = new_board[new_blank], new_board[blank]
                moves.append(PuzzleState(new_board, self, move, self.depth + 1))
        return moves

    def path(self):
        node, p = self, []
        while node:
            p.append((node.move, node.board, node.depth))
            node = node.parent
        return list(reversed(p))


def misplaced_tiles(state, goal):
    return sum(1 for i in range(9) if state.board[i] != 0 and state.board[i] != goal[i])
```

```
def manhattan_distance(state, goal):
    dist = 0
    for i in range(9):
        if state.board[i] != 0:
            r1, c1 = divmod(i, 3)
            r2, c2 = divmod(goal.index(state.board[i]), 3)
            dist += abs(r1 - r2) + abs(c1 - c2)
    return dist


def a_star(start, goal, heuristic):
    open_list = []
    closed_set = set()
    start_state = PuzzleState(start)
    start_state.cost = heuristic(start_state, goal)
    heapq.heappush(open_list, start_state)

    while open_list:
        current = heapq.heappop(open_list)

        if current.board == goal:
            return current.path()

        closed_set.add(tuple(current.board))

        for neighbor in current.get_moves():
            if tuple(neighbor.board) in closed_set:
                continue
            neighbor.cost = neighbor.depth + heuristic(neighbor, goal)
            heapq.heappush(open_list, neighbor)

    return None


def print_solution(solution):
    print("Solution steps:")
    for move, state, depth in solution:
        indent = "  " * depth  # indent based on depth for tree-like view
        if move == "":
            print(f"{indent}Start State (Depth {depth}):")
        else:
            print(f"{indent}Move {move} (Depth {depth}):")
        for i in range(0, 9, 3):
            print(indent + " ".join(str(x) if x != 0 else " " for x in state[i:i+3]))
        print()
```

```
if __name__ == "__main__":
    initial_state = [1, 2, 3,
                     4, 0, 6,
                     7, 5, 8]

    goal_state =    [1, 2, 3,
                     4, 5, 6,
                     7, 8, 0]
    print("A* with Misplaced Tiles Heuristic:\n")
    solution1 = a_star(initial_state, goal_state, misplaced_tiles)
    if solution1:
        print_solution(solution1)
    else:
        print("No solution found.")

    print("\nA* with Manhattan Distance Heuristic:\n")
    solution2 = a_star(initial_state, goal_state, manhattan_distance)
    if solution2:
        print_solution(solution2)
    else:
        print("No solution found.")
```

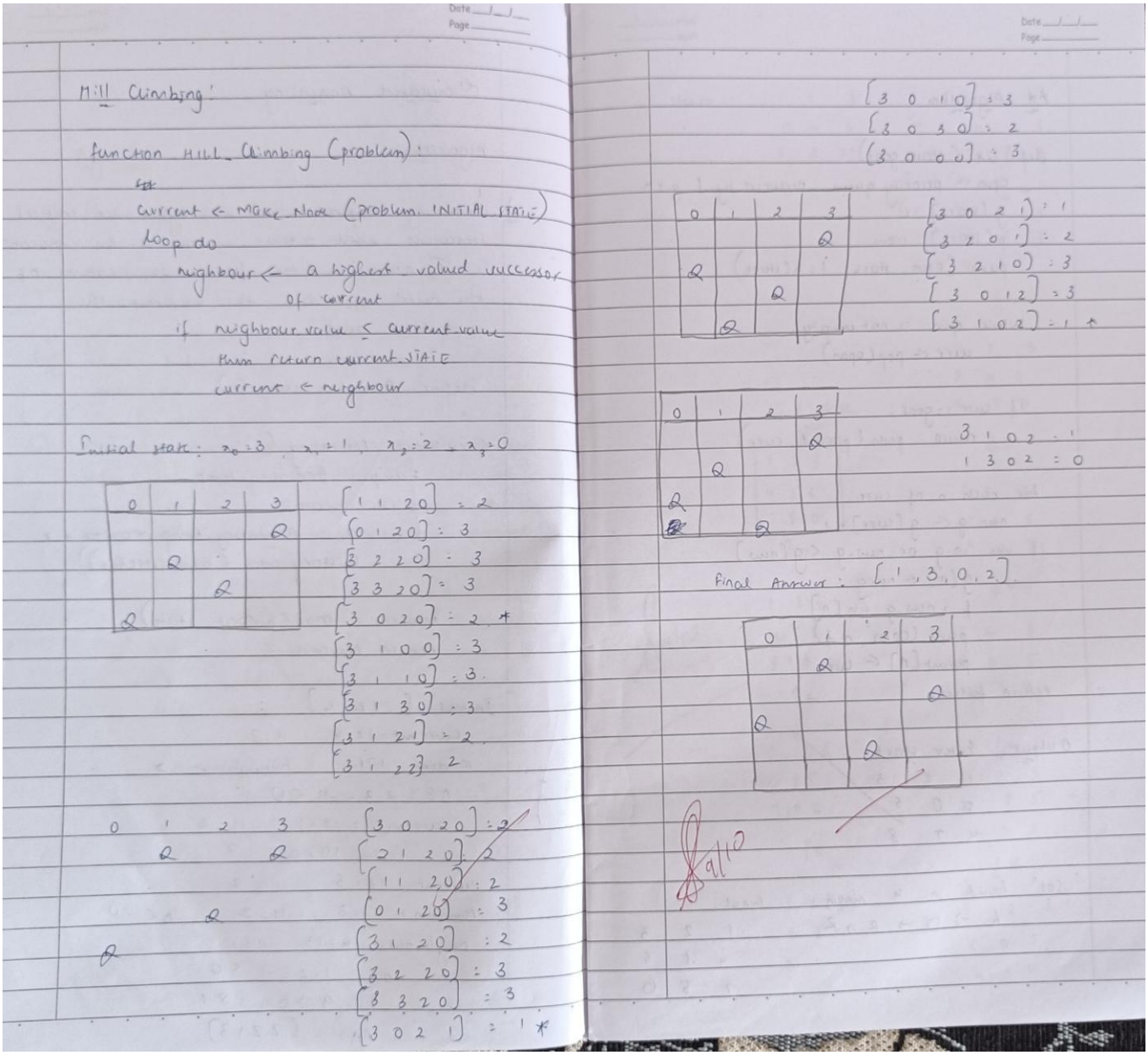
Output:

```
A* with Misplaced Tiles Heuristic:

Solution steps:
Start State (Depth 0):
1 2 3
4   6
7 5 8

  Move Down (Depth 1):
  1 2 3
  4 5 6
  7   8

    Move Right (Depth 2):
    1 2 3
    4 5 6
    7 8


A* with Manhattan Distance Heuristic:

Solution steps:
Start State (Depth 0):
1 2 3
4   6
7 5 8

  Move Down (Depth 1):
  1 2 3
  4 5 6
  7   8

    Move Right (Depth 2):
    1 2 3
    4 5 6
    7 8
```

Program 4:

Implement Hill Climbing search algorithm to solve N-Queens problem

Algorithm:

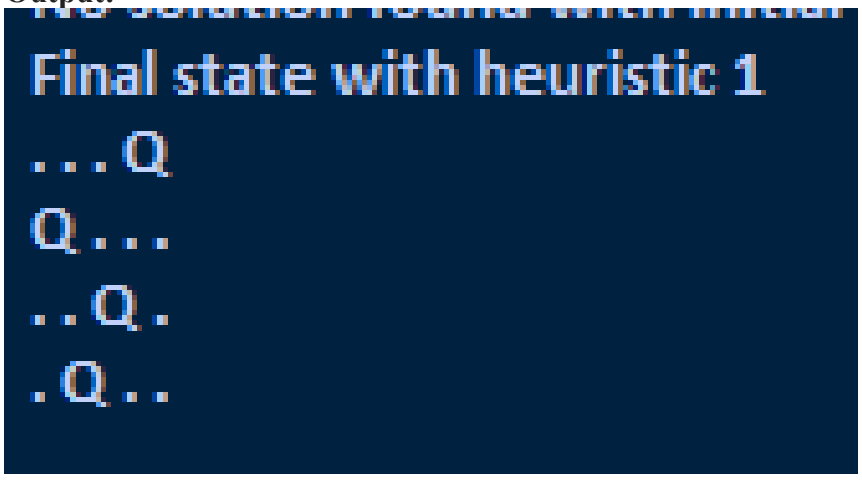


Code:

```
import random

def print_board(state):
    n = len(state)
    for row in range(n):
        line = ""
        for col in range(n):
            if state[row] == col:
                line += "Q "
            else:
                line += ". "
        print(line)
    print()

def heuristic(state):

    h = 0
    n = len(state)
    for i in range(n):
        for j in range(i + 1, n):
            if state[i] == state[j] or abs(state[i] - state[j]) == abs(i - j):
                h += 1
    return h

def get_neighbors(state):

    neighbors = []
    n = len(state)
    for row in range(n):
        for col in range(n):
            if col != state[row]:
                neighbor = state.copy()
                neighbor[row] = col
                neighbors.append(neighbor)
    return neighbors

def hill_climbing(initial_state):
    current = initial_state
    current_h = heuristic(current)

    while True:
        neighbors = get_neighbors(current)
        neighbor_h_pairs = [(heuristic(neighbor), neighbor) for neighbor in neighbors]

        # Find neighbor with minimum heuristic
        neighbor_h_pairs.sort(key=lambda x: x[0])
        best_h, best_neighbor = neighbor_h_pairs[0]
```

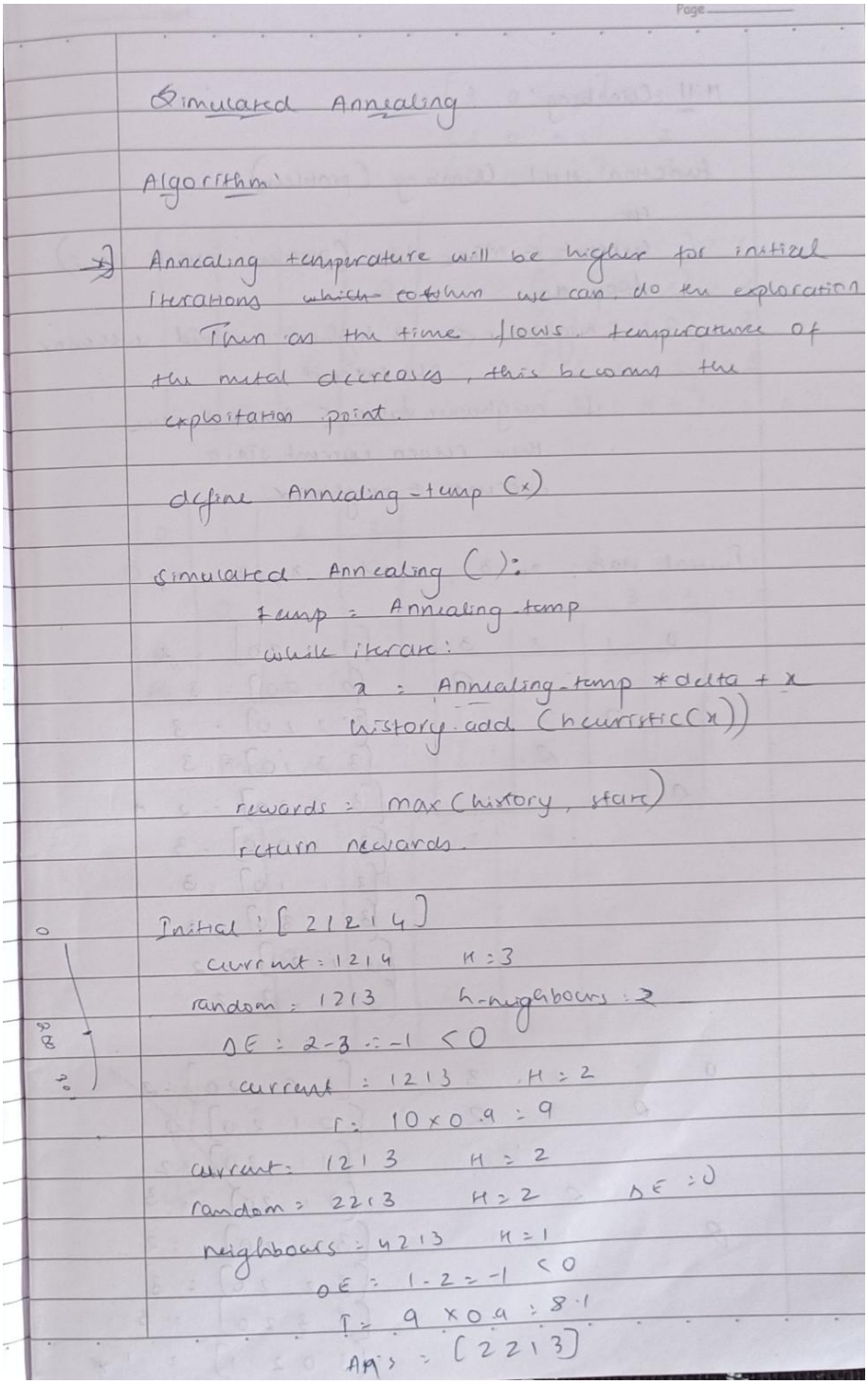
```
        # If no improvement, return current state
        if best_h >= current_h:
            return current, current_h

        current = best_neighbor
        current_h = best_h

def solve_n_queens(n=4):

    initial_state = [random.randint(0, n-1) for _ in range(n)]
    solution, h = hill_climbing(initial_state)

    if h == 0:
        print("Solution found:")
        print_board(solution)
    else:
        print("No solution found with initial state, try again.")
        print("Final state with heuristic", h)
        print_board(solution)

# Run the solver
solve_n_queens()
```

Output:



Program 5:
Simulated Annealing to Solve 8-Queens problem

Algorithm:



Simulated Annealing

Algorithm:

→ Annealing temperature will be higher for initial iterations which is to run we can do the exploration. Then as the time flows, temperatures of the metal decreases, this becomes the exploitation point.

define Annealing_temp (x)

Simulated Annealing ( ):
    Temp = Annealing_temp
    while iterate:
        x = Annealing_temp * delta + x
        history.add (heuristic(x))

    rewards = max (history, start)
    return rewards.

Initial : [2 1 2 1 4]
current : 1214 H : 3
random : 1213 h-neighbours : 2
ΔE : 2-3 : -1 < 0
current : 1213 H : 2
T : 10 x 0.9 : 9
current : 1213 H : 2
random : 2213 H : 2 ΔE : 0
neighbours : 4213 H : 1
ΔE : 1-2 = -1 < 0
T = 9 x 0.9 : 8.1
Ans : [2213]

Code:

```
import random
import math

def print_board(board):
    n = len(board)
    for i in range(n):
        row = ["Q" if board[i] == j else "." for j in range(n)]
        print(" ".join(row))
    print()

def calculate_cost(board):
    """Heuristic: number of pairs of queens attacking each other"""
    n = len(board)
    cost = 0
    for i in range(n):
        for j in range(i + 1, n):
            if board[i] == board[j] or abs(board[i] - board[j]) == abs(i - j):
                cost += 1
    return cost

def random_neighbor(board):
    """Generate a random neighboring board by moving one queen"""
    n = len(board)
    neighbor = list(board)
    row = random.randint(0, n - 1)
    col = random.randint(0, n - 1)
    neighbor[row] = col
    return neighbor

def simulated_annealing(n, initial_temp=100, cooling_rate=0.95, stopping_temp=1):
    current_board = [random.randint(0, n - 1) for _ in range(n)]
    current_cost = calculate_cost(current_board)
    temperature = initial_temp
    step = 1

    print("Initial Board:")
    print_board(current_board)
    print(f"Initial Cost: {current_cost}\n")

    while temperature > stopping_temp and current_cost > 0:
        neighbor = random_neighbor(current_board)
        neighbor_cost = calculate_cost(neighbor)
        delta = neighbor_cost - current_cost

        # Acceptance probability
        if delta < 0 or random.random() < math.exp(-delta / temperature):
            current_board = neighbor
            current_cost = neighbor_cost

        print(f"Step {step}: Temp={temperature:.3f}, Cost={current_cost}")
```

```
        step += 1
        temperature *= cooling_rate

    print("\nFinal Board:")
    print_board(current_board)
    print(f"Final Cost: {current_cost}")
    if current_cost == 0:
        print("Goal State Reached!")
    else:
        print("Terminated before reaching goal.")

# Run for 8-Queens
simulated_annealing(8)
```

Output:

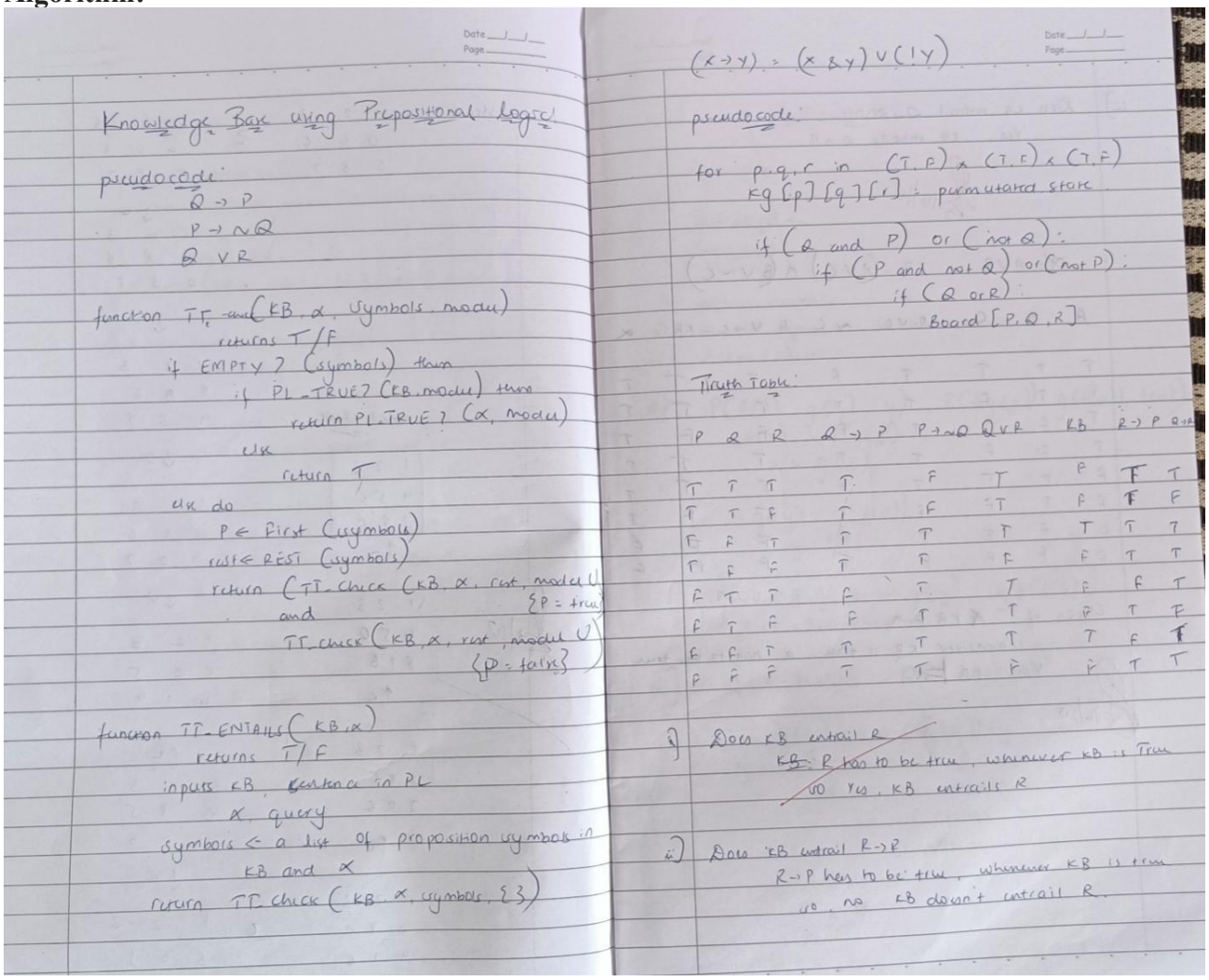
```
Initial Board:
...Q....
...Q....
......Q.
......Q.
.......Q
......Q.
......Q.
......Q.

Initial Cost: 15
```

```
Final Board:
Q.......
...Q....
.....Q..
Q.......
..Q.....
Q.......
......Q.
.Q......

Final Cost: 4
Terminated before reaching goal.
```

Program 6:

Create a knowledge base using propositional logic and show that the given query entails the knowledge base or not.

Algorithm:



Date __/__/__
Page

ii) Does KB entail $Q \rightarrow R$.

Yes, KB entails $Q \rightarrow R$.

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α: $A \vee B$ $\quad KB = (A \vee C) \wedge (B \vee \sim C)$

A	B	C	$A \vee C$	$\sim C$	$B \vee \sim C$	KB	α
T	T	T	T	F	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	F	F	F	T
T	F	F	T	T	T	T	T
F	T	T	T	F	T	T	T
F	T	F	F	T	T	F	T
F	F	T	T	F	F	F	F
F	F	F	F	T	F	F	F

Check if α KB $\models \alpha$

whenever KB is true, α has to be true

Yes KB $\models \alpha$.

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Code:

```
from itertools import product

def extract_symbols(expr):
    symbols = set()
    for c in expr:
        if c.isalpha():
            symbols.add(c)
    return sorted(symbols)

def replace_implications(expr):
    # If no implication, return as is
    if '=>' not in expr:
        return expr
    # Split on first occurrence of =>
    left, right = expr.split('=>', 1)
    left = left.strip()
    right = right.strip()
    # Replace implication by ((not (left)) or (right))
    new_expr = f"((not ({left})) or ({right}))"
    return new_expr

def eval_expr(expr, model):
    for sym, val in model.items():
        expr = expr.replace(sym, str(val))
    expr = expr.replace('~', ' not ')
    expr = expr.replace('&', ' and ')
    expr = expr.replace('|', ' or ')

    # Replace implication once
    expr = replace_implications(expr)

    expr = expr.replace('<=>', '==')

    return eval(expr)

def truth_table(KB_sentences):
    symbols = set()
    for s in KB_sentences:
        symbols |= set(extract_symbols(s))
    symbols = sorted(symbols)

    print("Truth Table:")
    header = symbols + KB_sentences
    print(" | ".join(f"{h:8}" for h in header))
    print("-" * (10 * (len(header))))

    models_where_KB_true = []
    for values in product([True, False], repeat=len(symbols)):
        model = dict(zip(symbols, values))
        KB_vals = [str(eval_expr(sentence, model)) for sentence in KB_sentences]
```

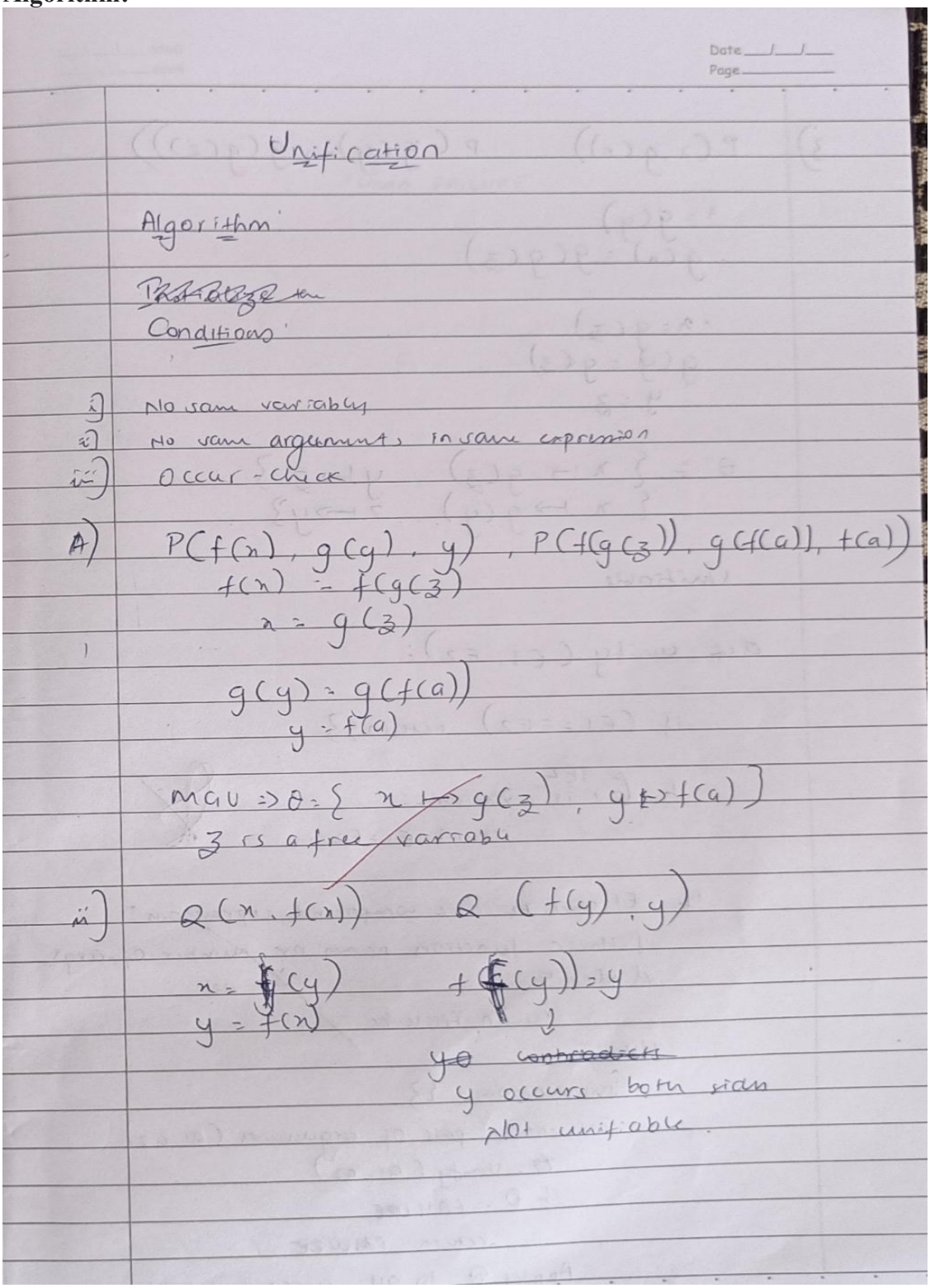
```
        row_vals = [str(model[s]) for s in symbols]
        print(" | ".join(f"{v:8}" for v in (row_vals + KB_vals)))
        if all(val == 'True' for val in KB_vals):
            models_where_KB_true.append(model)
    print("\nModels where KB is true:")
    for m in models_where_KB_true:
        print(m)
    return models_where_KB_true

def tt_entails(KB_sentences, query):
    symbols = set()
    for s in KB_sentences + [query]:
        symbols |= set(extract_symbols(s))
    symbols = sorted(symbols)

    for values in product([True, False], repeat=len(symbols)):
        model = dict(zip(symbols, values))
        if all(eval_expr(sentence, model) for sentence in KB_sentences):
            if not eval_expr(query, model):
                return False
    return True

# Your KB and queries
KB = [
    "Q => P",
    "P => ~Q",
    "Q | R"
]

queries = [
    "R",
    "R => P",
    "Q => R"
]

models = truth_table(KB)

for q in queries:
    print(f"\nDoes KB entail '{q}'? {tt_entails(KB, q)}")
```

Output:

```
---------- RESTART: C:/Users/student/Desktop/ai_week6.py ----------
Truth Table:
P        | Q        | R        | Q => P   | P => ~Q  | Q | R
-------------------------------------------------------------
True     | True     | True     | True     | False    | True
True     | True     | False    | True     | False    | True
True     | False    | True     | True     | True     | True
True     | False    | False    | True     | True     | False
False    | True     | True     | False    | True     | True
False    | True     | False    | False    | True     | True
False    | False    | True     | True     | True     | True
False    | False    | False    | True     | True     | False

Models where KB is true:
{'P': True, 'Q': False, 'R': True}
{'P': False, 'Q': False, 'R': True}

Does KB entail 'R'? True

Does KB entail 'R => P'? False

Does KB entail 'Q => R'? True
```

Program 7:
Implement unification in first order logic

Algorithm:



3) P(x, g(x)) P(g(y), g(g(z)))

x : g(y)
g(x) : g(g(z))

x : g(z)
g(y) : g(z)
y : z

θ : { x ↦ g(z), y ↦ z }
{ x ↦ g(y), z ↦ y }

Unifiable

def unify(E1, E2):

if (E1 == E2) return {}

if E1 / E2 is a variable:
return E1 : E2

if E1 and E2 are compound expressions:
if their function names or number of args differ:
return failure

else:

substitutions = {}
for each pair of arguments (a1, a2):
θ = unify(a1, a2)
if θ == FAILURE
return FAILURE
Apply θ to all current substitutions.
Add θ to substitutions

else:
return FAILURE

Code:

```
class Var:
    def __init__(self, name):
        self.name = name

    def __repr__(self):
        return self.name

    def __eq__(self, other):
        return isinstance(other, Var) and self.name == other.name

    def __hash__(self):
        return hash(self.name)

class Const:
    def __init__(self, name):
        self.name = name

    def __repr__(self):
        return self.name

    def __eq__(self, other):
        return isinstance(other, Const) and self.name == other.name

    def __hash__(self):
        return hash(self.name)

class Func:
    def __init__(self, name, args):
        self.name = name
        self.args = args  # list of terms

    def __repr__(self):
        return f"{self.name}({', '.join(map(str, self.args))})"

    def __eq__(self, other):
        return (isinstance(other, Func) and self.name == other.name and
                len(self.args) == len(other.args) and
                all(x == y for x, y in zip(self.args, other.args)))

    def __hash__(self):
        return hash((self.name, tuple(self.args)))

def is_variable(x):
    return isinstance(x, Var)

def is_compound(x):
    return isinstance(x, Func)

def occurs_check(var, term, subst):
    if var == term:
        return True
```

```
    elif is_variable(term) and term in subst:
        return occurs_check(var, subst[term], subst)
    elif is_compound(term):
        return any(occurs_check(var, arg, subst) for arg in term.args)
    else:
        return False

def unify(x, y, subst={}, depth=0):
    indent = "  " * depth  # for pretty printing nested calls
    if subst is None:
        return None
    elif x == y:
        # Terms are already identical
        print(f"{indent}Unify {x} and {y}: terms are identical, no substitution needed")
        return subst
    elif is_variable(x):
        return unify_var(x, y, subst, depth)
    elif is_variable(y):
        return unify_var(y, x, subst, depth)
    elif is_compound(x) and is_compound(y):
        if x.name != y.name or len(x.args) != len(y.args):
            print(f"{indent}Cannot unify different function symbols or different arity: {x} and {y}")
            return None
        for x_arg, y_arg in zip(x.args, y.args):
            subst = unify(x_arg, y_arg, subst, depth + 1)
            if subst is None:
                return None
        return subst
    else:
        print(f"{indent}Failed to unify {x} and {y}")
        return None

def unify_var(var, x, subst, depth):
    indent = "  " * depth
    if var in subst:
        print(f"{indent}Variable {var} already substituted with {subst[var]}, try to unify
{subst[var]} and {x}")
        return unify(subst[var], x, subst, depth + 1)
    elif is_variable(x) and x in subst:
        print(f"{indent}Variable {x} already substituted with {subst[x]}, try to unify {var} and
{subst[x]}")
        return unify(var, subst[x], subst, depth + 1)
    elif occurs_check(var, x, subst):
        print(f"{indent}Occurs check failed: {var} occurs in {x}")
        return None
    else:
        print(f"{indent}Substitute {var} with {x}")
        subst_copy = subst.copy()
        subst_copy[var] = x
        print(f"{indent}Current substitution: {subst_copy}")
        return subst_copy
```

```
if __name__ == "__main__":
    # Define terms:
    x = Var('x')
    y = Var('y')
    f1 = Func('f', [x, Const('a')])
    f2 = Func('f', [Const('b'), y])

    print("Start statement:")
    print(f"  Term 1: {f1}")
    print(f"  Term 2: {f2}")
    print("\nUnification steps:")

    result = unify(f1, f2, {})

    print("\nEnd state:")
    if result is None:
        print("  Unification failed.")
    else:
        print("  Final substitution:")
        for var, val in result.items():
            print(f"    {var} -> {val}")
```

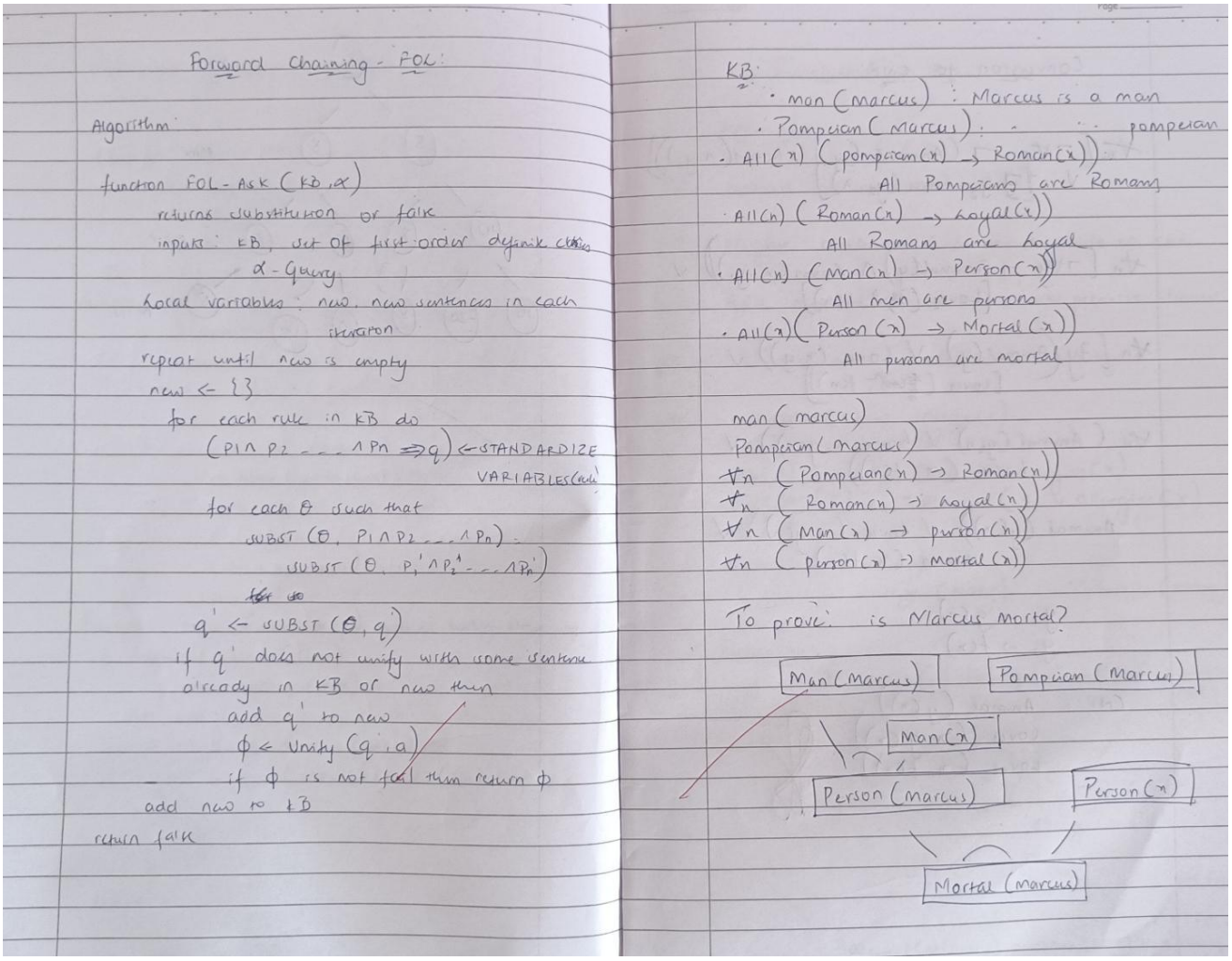
Output:

```
Start statement:
  Term 1: f(x, a)
  Term 2: f(b, y)

Unification steps:
  Substitute x with b
  Current substitution: {x: b}
  Substitute y with a
  Current substitution: {x: b, y: a}

End state:
  Final substitution:
    x -> b
    y -> a
```

Program 8:

Create a knowledge base consisting of first order logic statements and prove the given query using forward reasoning.

Algorithm:



Code:

```
class Predicate:
    def __init__(self, name, args):
        self.name = name
        self.args = args  # list of constants or variables

    def __repr__(self):
        return f"{self.name}({', '.join(map(str, self.args))})"

    def __eq__(self, other):
        return isinstance(other, Predicate) and self.name == other.name and self.args == other.args

    def __hash__(self):
        return hash((self.name, tuple(self.args)))

class Var:
    def __init__(self, name):
        self.name = name

    def __repr__(self):
        return self.name

    def __eq__(self, other):
        return isinstance(other, Var) and self.name == other.name

    def __hash__(self):
        return hash(self.name)

class Const:
    def __init__(self, name):
        self.name = name

    def __repr__(self):
        return self.name

    def __eq__(self, other):
        return isinstance(other, Const) and self.name == other.name

    def __hash__(self):
        return hash(self.name)

def is_variable(x):
    return isinstance(x, Var)

def unify(x, y, subst={}):
    if subst is None:
        return None
    elif x == y:
        return subst
    elif is_variable(x):
```

```
        return unify_var(x, y, subst)
    elif is_variable(y):
        return unify_var(y, x, subst)
    elif isinstance(x, Predicate) and isinstance(y, Predicate):
        if x.name != y.name or len(x.args) != len(y.args):
            return None
        for a, b in zip(x.args, y.args):
            subst = unify(a, b, subst)
            if subst is None:
                return None
        return subst
    else:
        return None

def unify_var(var, x, subst):
    if var in subst:
        return unify(subst[var], x, subst)
    elif is_variable(x) and x in subst:
        return unify(var, subst[x], subst)
    elif occurs_check(var, x, subst):
        return None
    else:
        subst_copy = subst.copy()
        subst_copy[var] = x
        return subst_copy

def occurs_check(var, x, subst):
    if var == x:
        return True
    elif is_variable(x) and x in subst:
        return occurs_check(var, subst[x], subst)
    elif isinstance(x, Predicate):
        return any(occurs_check(var, arg, subst) for arg in x.args)
    else:
        return False

def substitute(predicate, subst):
    new_args = []
    for arg in predicate.args:
        val = arg
        while is_variable(val) and val in subst:
            val = subst[val]
        new_args.append(val)
    return Predicate(predicate.name, new_args)

class Rule:
    def __init__(self, premises, conclusion):
        self.premises = premises  # list of Predicate
        self.conclusion = conclusion  # Predicate

    def __repr__(self):
        return f"{' ∧ '.join(map(str, self.premises))} => {self.conclusion}"
```

```
def forward_chain(kb_facts, kb_rules, query):
    inferred = set(kb_facts)
    print("Initial Facts:")
    for f in inferred:
        print(f"  {f}")
    print("\nStarting inference steps:\n")

    new_inferred = True

    while new_inferred:
        new_inferred = False
        for rule in kb_rules:
            possible_substs = [{}]  # substitutions for premises

            for premise in rule.premises:
                temp_substs = []
                for fact in inferred:
                    for subst in possible_substs:
                        subst_try = unify(premise, fact, subst)
                        if subst_try is not None:
                            temp_substs.append(subst_try)
                possible_substs = temp_substs

            for subst in possible_substs:
                concluded_fact = substitute(rule.conclusion, subst)
                if concluded_fact not in inferred:
                    print(f"Inferred: {concluded_fact} from rule {rule} using substitution {subst}")
                    inferred.add(concluded_fact)
                    new_inferred = True
                    if unify(concluded_fact, query) is not None:
                        print(f"\nQuery {query} proved!")
                        return True

    print(f"\nQuery {query} not proved.")
    return False

if __name__ == "__main__":
    a = Const('a')
    b = Const('b')
    c = Const('c')
    x = Var('x')
    y = Var('y')
    z = Var('z')

    kb_facts = {
        Predicate('Parent', [a, b]),
        Predicate('Parent', [b, c]),
    }

    kb_rules = [
        Rule([Predicate('Parent', [x, y])], Predicate('Ancestor', [x, y])),
```

```
    Rule([Predicate('Parent', [x, y]), Predicate('Ancestor', [y, z])], Predicate('Ancestor', [x, z])),
]

query = Predicate('Ancestor', [a, c])

print("Running forward chaining...\n")
forward_chain(kb_facts, kb_rules, query)
```

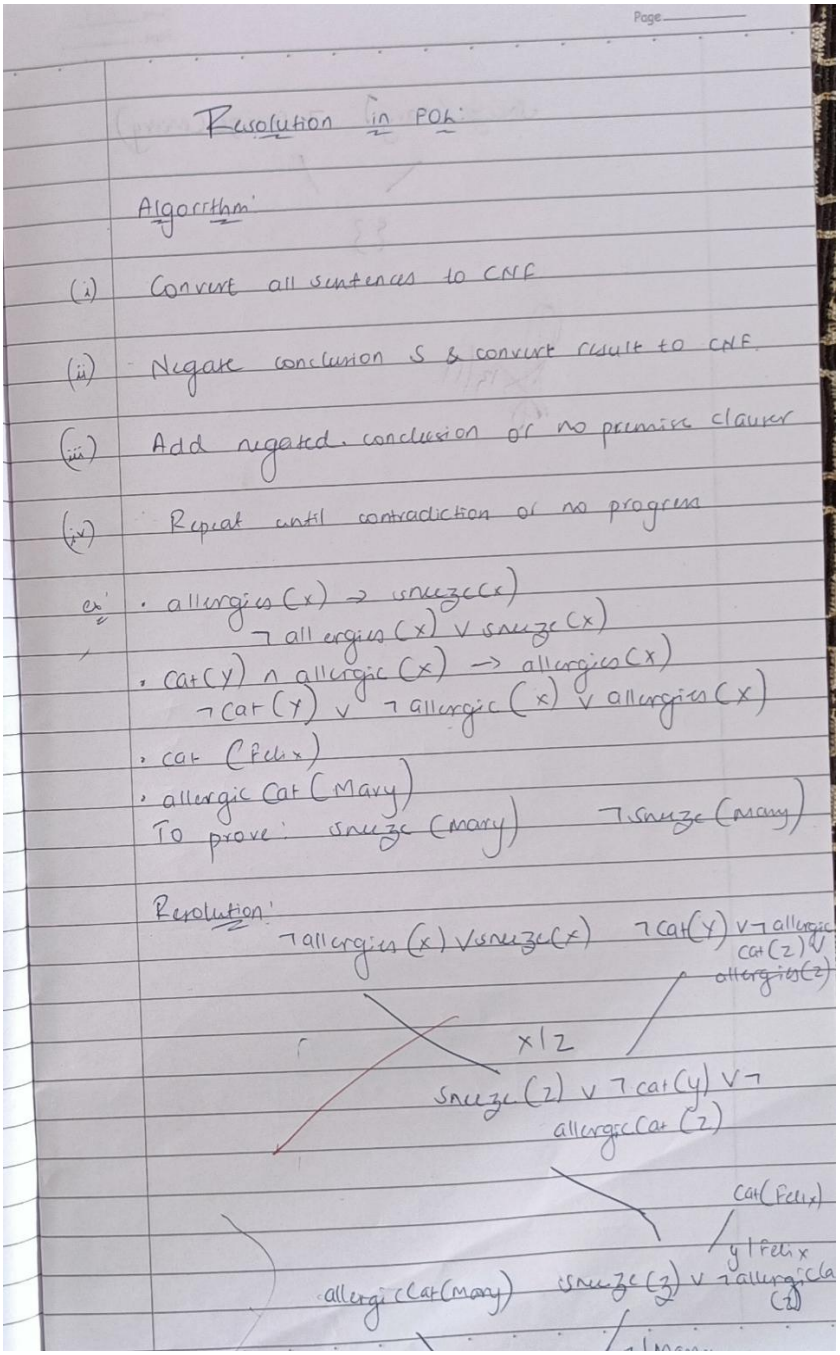
Output:

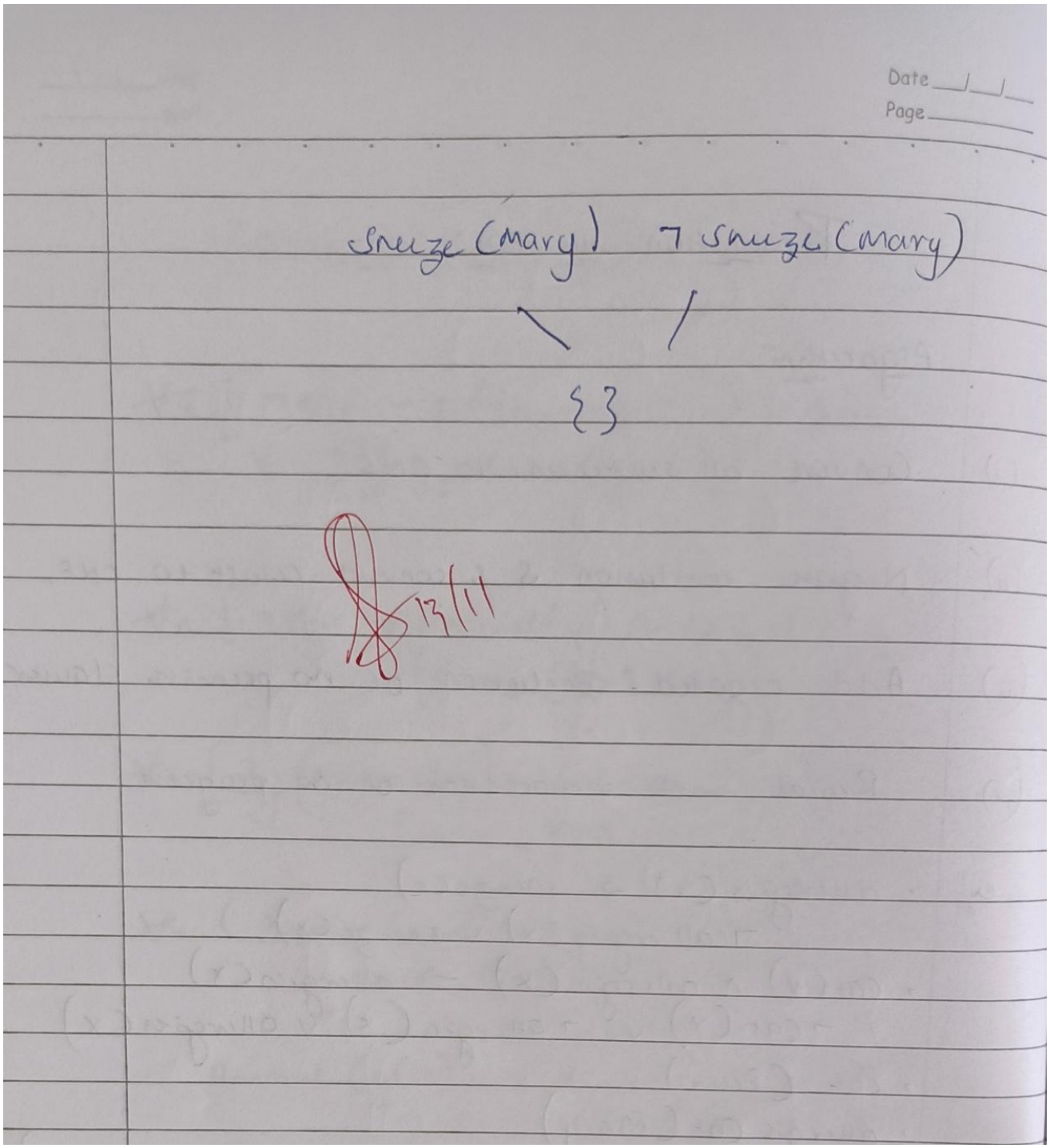
```
========== RESTART: C:/Users/STUDENT/Desktop/AI_11
Running forward chaining...

Initial Facts:
  Parent(a, b)
  Parent(b, c)

Starting inference steps:

Inferred: Ancestor(a, b) from rule Parent(x, y) => Ancestor(x, y) using substitution {x: a, y: b}
Inferred: Ancestor(b, c) from rule Parent(x, y) => Ancestor(x, y) using substitution {x: b, y: c}
Inferred: Ancestor(a, c) from rule Parent(x, y) ∧ Ancestor(y, z) => Ancestor(x, z) using substitution {x: a, y: b, z: c}

Query Ancestor(a, c) proved!
```

Program 9:

Create a knowledge base consisting of first order logic statements and prove the given query using Resolution

Algorithm:




Date
Page
sneeze(mary) ⊢ sneeze(mary)
{}
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Code:

```
from copy import deepcopy

# ----------------------------
# Basic utilities
# ----------------------------

def is_variable(x):
    return isinstance(x, str) and x[0].islower()

def substitute(subst, expr):
    """Substitute variables in expr according to subst mapping."""
    if isinstance(expr, str):
        return subst.get(expr, expr)
    return (expr[0],) + tuple(substitute(subst, a) for a in expr[1:])

def unify(x, y, subst=None):
    """Unify two literals, returning substitution if possible."""
    if subst is None:
        subst = {}
    if x == y:
        return subst
    if is_variable(x):
        return unify_var(x, y, subst)
    if is_variable(y):
        return unify_var(y, x, subst)
    if isinstance(x, tuple) and isinstance(y, tuple) and x[0] == y[0] and len(x) == len(y):
        for a, b in zip(x[1:], y[1:]):
            subst = unify(substitute(subst, a), substitute(subst, b), subst)
            if subst is None:
                return None
        return subst
    return None

def unify_var(var, x, subst):
    if var in subst:
        return unify(subst[var], x, subst)
    elif x in subst:
        return unify(var, subst[x], subst)
    elif occurs_check(var, x, subst):
        return None
    else:
        subst[var] = x
        return subst

def occurs_check(var, x, subst):
    """Prevent circular substitutions."""
    if var == x:
        return True
    elif isinstance(x, tuple):
        return any(occurs_check(var, arg, subst) for arg in x[1:])
```

```
    elif is_variable(x) and x in subst:
        return occurs_check(var, subst[x], subst)
    return False

def negate(lit):
    return ('~' + lit[0][1:],) + lit[1:] if lit[0].startswith('~') else ('~' + lit[0],) + lit[1:]

# ----------------------------
# Resolution
# ----------------------------

def resolve(ci, cj):
    """Return resolvents between two clauses."""
    resolvents = []
    for li in ci:
        for lj in cj:
            if li[0].startswith('~') != lj[0].startswith('~') and li[0].lstrip('~') == lj[0].lstrip('~'):
                subst = unify(li, negate(lj))
                if subst is not None:
                    new_clause = set(substitute(subst, l) for l in ci.union(cj) if l != li and l != lj)
                    resolvents.append(frozenset(new_clause))
    return resolvents

def fol_resolution(kb, query):
    clauses = deepcopy(kb)
    clauses.append({negate(query)})
    print("Initial clauses:")
    for c in clauses:
        print("  ", c)
    print("\nResolving...\n")

    new = set()
    while True:
        pairs = [(clauses[i], clauses[j]) for i in range(len(clauses)) for j in range(i + 1, len(clauses))]
        for (ci, cj) in pairs:
            for resolvent in resolve(ci, cj):
                if not resolvent:
                    print("Derived empty clause -> contradiction found ☑")
                    return True
                new.add(resolvent)
        if new.issubset(set(map(frozenset, clauses))):
            print("No new clauses can be added ✗")
            return False
        for c in new:
            if c not in clauses:
                clauses.append(set(c))
                print("New clause:", c)

# ----------------------------
# Example KB (CNF)
# ----------------------------
```

```
KB = [
    {('~Food', 'x'), ('Likes', 'John', 'x')},
    {('Food', 'Apple')},
    {('Food', 'Vegetable')},
    {('~Eats', 'x', 'y'), ('Killed', 'x'), ('Food', 'y')},
    {('Eats', 'Anil', 'Peanut')},
    {('Alive', 'Anil')},
    {('~Eats', 'Anil', 'y'), ('Eats', 'Harry', 'y')},
    {('~Alive', 'x'), ('~Killed', 'x')},
    {('Killed', 'x'), ('Alive', 'x')}
]

query = ('Likes', 'John', 'Peanut')

# ----------------------------
# Run Resolution
# ----------------------------

proved = fol_resolution(KB, query)

if proved:
    print("\n☑ Proven: John likes peanuts.")
else:
    print("\n❌ Could not prove John likes peanuts.")
```

Output:

```
Initial clauses:
   {('Likes', 'John', 'x'), ('~Food', 'x')}
   {('Food', 'Apple')}
   {('Food', 'Vegetable')}
   {('Food', 'y'), ('~Eats', 'x', 'y'), ('Killed', 'x')}
   {('Eats', 'Anil', 'Peanut')}
   {('Alive', 'Anil')}
   {('~Eats', 'Anil', 'y'), ('Eats', 'Harry', 'y')}
   {('~Alive', 'x'), ('~Killed', 'x')}
   {('Alive', 'x'), ('Killed', 'x')}
   {('~Likes', 'John', 'Peanut')}

Resolving...

New clause: frozenset({('Killed', 'Harry'), ('Food', 'y'), ('~Eats', 'Anil', 'y')})
New clause: frozenset({('Killed', 'Anil'), ('Food', 'Peanut')})
New clause: frozenset({('Killed', 'x'), ('~Killed', 'x')})
New clause: frozenset({('Alive', 'x'), ('~Alive', 'x')})
New clause: frozenset({('Likes', 'John', 'Apple')})
New clause: frozenset({('Likes', 'John', 'Vegetable')})
New clause: frozenset({('~Eats', 'y', 'y'), ('Killed', 'y'), ('Likes', 'John', 'y')})
New clause: frozenset({('~Killed', 'x')})
New clause: frozenset({('Likes', 'John', 'Peanut'), ('Killed', 'Anil')})
New clause: frozenset({('Killed', 'Peanut'), ('~Eats', 'Peanut', 'Peanut')})
New clause: frozenset({('~Eats', 'y', 'y'), ('~Alive', 'y'), ('Likes', 'John', 'y')})
New clause: frozenset({('~Alive', 'Anil'), ('Food', 'Peanut')})
New clause: frozenset({('~Eats', 'Anil', 'y'), ('Killed', 'Harry'), ('Likes', 'John', 'y')})
New clause: frozenset({('~Alive', 'x')})
New clause: frozenset({('~Alive', 'Harry'), ('~Eats', 'Anil', 'y'), ('Food', 'y')})
New clause: frozenset({('Likes', 'John', 'Peanut'), ('~Alive', 'Anil')})
New clause: frozenset({('~Eats', 'Anil', 'Peanut'), ('Killed', 'Harry')})
New clause: frozenset({('Likes', 'John', 'Peanut')})
New clause: frozenset({('Killed', 'Anil')})
New clause: frozenset({('~Eats', 'Peanut', 'Peanut')})
New clause: frozenset({('~Eats', 'Anil', 'y'), ('Likes', 'John', 'y')})
New clause: frozenset({('~Alive', 'Peanut'), ('~Eats', 'Peanut', 'Peanut')})
New clause: frozenset({('~Eats', 'Anil', 'y'), ('~Alive', 'Harry'), ('Likes', 'John', 'y')})
Derived empty clause -> contradiction found ☑

☑ Proven: John likes peanuts.
```

Program 10:
Implement Alpha-Beta Pruning.

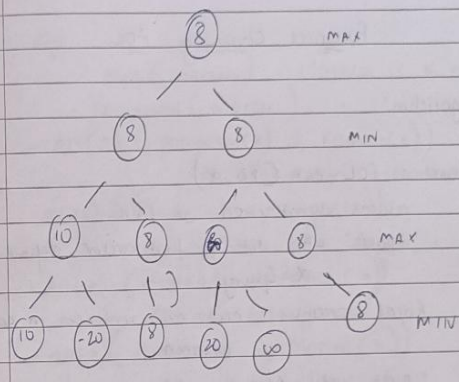
Algorithm:

Min-max algorithm, α β pruning:

```
function ALPHA-BETA-SEARCH (state)
    returns action
    v ← MAXV(state, -∞, +∞)
    return action in ACTIONS (state) with v

function MAXV (state, α, β) returns utility value
    if TERMINAL-TEST (state)
        return UTILITY (state)
    v ← -∞

    for each a in ACTIONS (state) do
        v ← MAX (v, MINV (RESULT (s, a) α, β))
        if v ≥ β
            return v
        α ← MAX (α, v)
        return v
function MINV (state, α, β) returns utility state
    if terminal-state (state)
        return utility (state)
    v ← +∞
    for each a in actions (state) do
        v ← min ( v, MAXV (RESULT (s,a) α, β))
        if v ≤ α
            return v
        β ← MIN (β, v)
    return v
```



Code:

```
import math

# ============================================================
# Define a simple binary game tree
# ============================================================

class Node:
    def __init__(self, value=None, children=None):
        self.value = value          # Leaf value (for terminal nodes)
        self.children = children or []  # List of child nodes


# ============================================================
# Alpha–Beta Pruning Algorithm
# ============================================================

def alpha_beta(node, depth, alpha, beta, maximizing_player):
    """
    Alpha–Beta pruning on an explicit tree structure.
    """
    # Base case: leaf node
    if depth == 0 or not node.children:
        return node.value

    if maximizing_player:
        max_eval = -math.inf
        for child in node.children:
            eval = alpha_beta(child, depth - 1, alpha, beta, False)
            max_eval = max(max_eval, eval)
            alpha = max(alpha, eval)
            if beta <= alpha:
                break  # prune
        return max_eval
    else:
        min_eval = math.inf
        for child in node.children:
            eval = alpha_beta(child, depth - 1, alpha, beta, True)
            min_eval = min(min_eval, eval)
            beta = min(beta, eval)
            if beta <= alpha:
                break  # prune
        return min_eval


# ============================================================
# Trace version (to see pruning steps)
# ============================================================

def alpha_beta_trace(node, depth, alpha, beta, maximizing_player, indent=""):
    if depth == 0 or not node.children:
```

```
        print(f"{indent}Leaf → value = {node.value}")
        return node.value

    if maximizing_player:
        print(f"{indent}MAX node, α={alpha}, β={beta}")
        value = -math.inf
        for i, child in enumerate(node.children):
            print(f"{indent}→ Explore MIN child {i}")
            eval = alpha_beta_trace(child, depth - 1, alpha, beta, False, indent + "  ")
            value = max(value, eval)
            alpha = max(alpha, value)
            print(f"{indent}MAX updated: value={value}, α={alpha}, β={beta}")
            if beta <= alpha:
                print(f"{indent}PRUNE remaining children (β ≤ α)\n")
                break
        return value
    else:
        print(f"{indent}MIN node, α={alpha}, β={beta}")
        value = math.inf
        for i, child in enumerate(node.children):
            print(f"{indent}→ Explore MAX child {i}")
            eval = alpha_beta_trace(child, depth - 1, alpha, beta, True, indent + "  ")
            value = min(value, eval)
            beta = min(beta, value)
            print(f"{indent}MIN updated: value={value}, α={alpha}, β={beta}")
            if beta <= alpha:
                print(f"{indent}PRUNE remaining children (β ≤ α)\n")
                break
        return value
# ============================================================
# Example Tree (Depth 3, 8 leaf nodes)
# ============================================================
# Leaves
leaf_nodes = [Node(3), Node(5), Node(6), Node(9), Node(1), Node(2), Node(0), Node(-1)]

# Level 2 (MIN nodes)
level2 = [
    Node(children=[leaf_nodes[0], leaf_nodes[1]]),
    Node(children=[leaf_nodes[2], leaf_nodes[3]]),
    Node(children=[leaf_nodes[4], leaf_nodes[5]]),
    Node(children=[leaf_nodes[6], leaf_nodes[7]])
]

# Level 1 (MAX nodes)
level1 = [
    Node(children=[level2[0], level2[1]]),
    Node(children=[level2[2], level2[3]])
]

# Root (MIN node)
root = Node(children=[level1[0], level1[1]])
```

```
# ============================================================
# Run the algorithm
# ============================================================
if __name__ == "__main__":
    print("========== Alpha–Beta Pruning (No Trace) ==========")
    optimal_value = alpha_beta(root, 3, -math.inf, math.inf, True)
    print(f"\n☑ Optimal value found: {optimal_value}")

    print("\n========== Alpha–Beta Pruning (Trace) ==========\n")
    traced_value = alpha_beta_trace(root, 3, -math.inf, math.inf, True)
    print(f"\n☑ Final optimal value (trace) = {traced_value}")
```

Output:

```
========== Alpha–Beta Pruning (No Trace) ==========

☑ Optimal value found: 5

========== Alpha–Beta Pruning (Trace) ==========

MAX node, α=-inf, β=inf
→ Explore MIN child 0
  MIN node, α=-inf, β=inf
  → Explore MAX child 0
    MAX node, α=-inf, β=inf
    → Explore MIN child 0
      Leaf → value = 3
    MAX updated: value=3, α=3, β=inf
    → Explore MIN child 1
      Leaf → value = 5
    MAX updated: value=5, α=5, β=inf
  MIN updated: value=5, α=-inf, β=5
  → Explore MAX child 1
    MAX node, α=-inf, β=5
    → Explore MIN child 0
      Leaf → value = 6
    MAX updated: value=6, α=6, β=5
    PRUNE remaining children (β ≤ α)

  MIN updated: value=5, α=-inf, β=5
MAX updated: value=5, α=5, β=inf
→ Explore MIN child 1
  MIN node, α=5, β=inf
  → Explore MAX child 0
    MAX node, α=5, β=inf
    → Explore MIN child 0
      Leaf → value = 1
    MAX updated: value=1, α=5, β=inf
    → Explore MIN child 1
      Leaf → value = 2
    MAX updated: value=2, α=5, β=inf
  MIN updated: value=2, α=5, β=2
  PRUNE remaining children (β ≤ α)

MAX updated: value=5, α=5, β=inf

☑ Final optimal value (trace) = 5
```